Weikun Tian, Ph.D.

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Profile / Highlights

- Postdoctoral researcher / Marie Skłodowska-Curie Actions (MSCA) Fellow at the Max-Planck Institute of Quantum Optics (MPQ), advised by Prof. Immanuel Bloch;
- Experienced in building cold-atom and cold-molecule experimental platforms and key subsystems from scratch, including laser frequency stabilization and phase locking, optical tweezer arrays, laser auto-alignment, and microwave systems;
- Built a 225-atom optical tweezer array platform from scratch during PhD studies and proposed a parallel assembly algorithm (Physical Review Applied 2023; Featured in Physics);
- First observed Fermi-surface deformation in a bulk gas of fermionic polar molecules, advancing studies of many-body physics with strong dipolar interactions.

Education

B.S. **Department of Physics, Zhejiang University, China** 2016.09–2020.06

Ph.D. **Centre for Quantum Technologies, National University of Singapore (NUS)** 2020.08–2024.03

PhD Thesis: Parallel assembly of defect-free Rydberg atom arrays for quantum simulation

Research Experience

Department of Physics, Zhejiang University | Undergraduate Research Assistant 2018.04–2018.08
Advisor: Prof. Bo Yan (Professor and Doctoral Supervisor, Department of Physics, Zhejiang University)
Project: Construction of a high-stability current controller

- Completed the full workflow from design to debugging of a high-stability current controller, achieving current instability below 10^{-6} and meeting the power-supply requirements of precision experiments;
- Developed systematic methods for analyzing and validating analog circuits and feedback control, establishing a reusable workflow for circuit debugging and error tracing.

Department of Physics, Zhejiang University | Undergraduate Research Assistant 2018.10–2019.05
Advisor: Prof. Bo Yan (Professor and Doctoral Supervisor, Department of Physics, Zhejiang University)
Project: Laser frequency-stabilization system based on DAVLL

- Designed and implemented a permanent-magnet scheme for generating uniform magnetic fields inside a vapor cell, enabling Zeeman splitting and DAVLL frequency locking;
- Stabilized an 852 nm laser to within 2 MHz, significantly improving the long-term stability and reproducibility of the experimental system.

Centre for Quantum Technologies, National University of Singapore | PhD Student 2020.08–2024.03
Advisor: Asst. Prof. Huanqian Loh (President's Assistant Professor, Department of Physics, NUS; Principal Investigator, CQT)
Project: Quantum simulation with two-dimensional triangular atom arrays

- Built a large-scale two-dimensional triangular-lattice atom-array platform from scratch, completing the optical, control, feedback, and calibration loops, and realized preparation of arrays with up to 225 atoms at a 33% success probability;
- Independently built key modules of the Rydberg laser system, including cavity locking, automatic alignment, and power/pointing stabilization, enabling homogeneous Rydberg excitation of atom arrays and stable quantum simulation experiments;
- Observed quantum thermalization restricted by Hilbert-space fragmentation in a one-dimensional Rydberg-array quantum simulator, providing experimental evidence for mechanisms of nonequilibrium quantum dynamics.

Max-Planck Institute of Quantum Optics | Postdoctoral Researcher / Marie Skłodowska-Curie Fellow 2024.05–Present
Advisor: Prof. Immanuel Bloch (Scientific Director, Max-Planck Institute of Quantum Optics; 2025 Micius Quantum Prize Laureate; 2022 Clarivate Citation Laureate)
Project: p-wave superfluidity and BCS–BEC crossover of ultracold polar molecules

- First observed Fermi-surface deformation in a bulk gas of fermionic polar molecules, realizing the first experimental observation of a many-body effect in this system;
- Designed an upgraded high-power, low-noise microwave system for molecular microwave shielding, improving stable control under strong dipolar interactions;
- Completed the design of the laser system for trapping molecules in an optical lattice, establishing optical infrastructure for future experiments in strongly correlated and superfluid regimes.

PUBLICATIONS

- S. Biswas, S. Eppelt, **W. Tian**, W. Zhang, F. Deng, C. Frank, T. Shi, I. Bloch, and X.-Y. Luo, "Controlled symmetry breaking of the Fermi surface in ultracold polar molecules," arXiv:2602.22447 (2026).
  Reported interaction-induced, controlled deformation of the Fermi surface in a degenerate gas of $^{23}Na^{40}K$ molecules. Demonstrated tunable symmetry breaking from axial U(1) to biaxial C_2 symmetry under microwave shielding.
- L. Zhao, P. R. Datla, **W. Tian**, M. M. Aliyu, and H. Loh, "Observation of quantum thermalization restricted to Hilbert space fragments and Z2k Scars," *Physical Review X* **15**, 011035 (2025).
  First observation of quantum thermalization restricted by Hilbert-space fragmentation, providing a new perspective for understanding thermalization in quantum systems.
- **W. Tian**, W. J. Wee, A. Qu, B. J. M. Lim, P. R. Datla, V. P. W. Koh and H. Loh, "Parallel assembly of arbitrary defect-free atom arrays with a multi-tweezer algorithm," *Physical Review Applied* **19**, 034048 (2023). Physics
  Proposed and experimentally validated a multi-tweezer parallel assembly algorithm for preparing arbitrary defect-free neutral atom arrays; realized defect-free arrays with up to 225 atoms;
  Featured in Physics.

CONFERENCES AND INVITED TALKS

- **53rd Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP)**
  Orlando (USA), 2022.05.30, Online Talk
  "Triangular arrays of hundreds of Rb atoms for quantum simulation"
- **53rd Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP)**
  Orlando (USA), 2022.05.30, Poster
  "Towards quantum many-body physics in two-dimensional triangular atom arrays"
- **Gordon Research Conference on Quantum Science**
  Easton (USA), 2022.07.24, Poster
  "Defect-free atom arrays in triangle-based geometries for quantum simulation"
- **The International Conference on Laser Spectroscopy 2023**
  Estes Park (USA), 2023.06.25, Poster
  "Geometrically frustrated Rb atom arrays in Rydberg states for quantum many-body simulation"
- **Invited Talk at Zhejiang University**
  Hangzhou (China), 2023.12.28, Invited Talk
  "Geometrically frustrated Rb atom arrays in Rydberg states for quantum many-body simulation"
- **Physical Review Journal Club**
  Online, 2024.12.18, Invited Talk
  "Parallel Assembly of Arbitrary Defect-Free Atom Arrays with a Multitweezer Algorithm"
- **Haikou Winter School: Fundamentals and Frontiers of Ultracold Atoms and Molecules**
  Haikou (China), 2025.12.04, Invited Talk
  "Fermi-surface deformation of microwave-shielded molecules"

SCHOLARSHIPS AND HONORS

- Third-Class Scholarship of Zhejiang University (ranked 11/74 in the program) 2018.10
- CQT PhD Scholarship 2020.08
- European Union Marie Skłodowska-Curie Actions (MSCA) Fellow 2025.05

SKILLS

- Experimental systems: optical tweezer arrays, Rydberg excitation, laser frequency stabilization, laser auto-alignment, microwave systems, vacuum systems, and experimental control;
- Programming languages and software: Python (data analysis and experiment automation), MATLAB (data analysis), C;
- Engineering design: Eagle CAD, SolidWorks, Fusion 360 for integrated mechanical, electronic, and optical design;
- Languages: Chinese (native), English (fluent).